Qwen-Image diffsynth ControlNet canny / depth / inpaint

A measured protection set, selected on the patch's own output. One layer carries 90.9 % of the measurable contribution.

CHECKPOINT	`models/model_patches/qwen_image_canny_diffsynth_controlnet.safetensors`
PACK	`qwen_image_canny_diffsynth_controlnet_convrot_int8.safetensors`
SOURCE	2161.8 MB · BF16 · 362 tensors
PACK	1083.3 MB · −49.9 % · 120 INT8 + 1 kept
SHA256	`C2899255252B81FD64EF2FAE0FBFD86E1496BA04CF10CC95FE45679120F1C996`
GATE	cos mean ≥ 0.99 · cos min ≥ 0.98 · finite · **PASS**

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1 · What the score is

The score is **not** a

Two choices the runtime forces

static weight statistic (kurtosis, absmax, outlier ratio). It is the **observed change of the model's own output** when a layer is perturbed — a behavioural measurement, taken in the frame the runtime uses.

For block `i` of the 60 control blocks:

```
cos_i =
1 - cos_
```

A pack's score is the **mean over blocks** of `1 -`

The observable is a list of 60 residuals, not one vector. The apply node does `img[:, :n] += control_block(img[:, :n], cond, i) * strength` per block, so the 60 residuals are independent. Concatenated, the largest block's norm decides the number and hides the rest.

The arithmetic is float64. The residuals run to ~3e5, so a float32 dot product over 47,185,920 elements accumulates ~1e-3 of rounding noise — larger than the signal (`img_in` is 1.12e-05, the smallest layer 2.6e-09). The first float32 run returned every value negative.

> The gate: `cos.mean ≥ 0.99` and `cos.min ≥ 0.98` over the fixed 256-item set, seed 42, finite output. Values are quoted to 10 decimals because candidates differ at 1e-07.

`cos_i`, then averaged over items. The reported `cos.mean` / `cos.min` are `1 − (the mean error)` and `1 − (the worst item's error)`.

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2 · This is a model patch, and it needs the HSWQ loader

The file name says "controlnet" but the runtime object is a **model patch**. ComfyUI dispatches it on `controlnet_blocks.0.y_rms.weight` into `QwenImageBlockWiseControlNet`: 60 blocks

Load it with the **HSWQ Model Patch Loader** (`HSWQModelPatchLoaderCustom`). It detects the `int8_tensorwise` `comfy_quant` stamps and builds the patch with `mixed_precision_ops`: weights stay INT8, and the forward uses the comfy-kitchen `int8_linear` kernel with online ConvRot rotation.

```
HSWQModelPatchLoaderCustom.load_model_patch(name, cpu_offload)
  → ModelPatcher            (the module is patcher.model)
  → QwenImageBlockWiseControlNet, 120 INT8, img_in kept
```

of `[x_rms, y_rms, input_proj, gelu, output_proj]` plus one shared `img_in`.

ComfyUI's stock `ModelPatchLoader` does not recognise the INT8 layout. It builds the patch with `manual_cast`, which reads the pack's INT8 tensors as plain weights.

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3 · The model

THE CHECKPOINT, MEASURED

ITEM	MEASURED

ITEM	MEASURED
the source	2161.8 MB · BF16 · 362 tensors
the 2D weights	**121**, all ConvRot-eligible
the layer families	`img_in` + `controlnet_blocks.{0..59}` (`input_proj` / `output_proj` ; `x_rms` / `y_rms` are 1D norms)
the group size	256 (all 121)

Unlike the Z-Image Fun ControlNet patch, **no key conversion happens on load** (only the Z-Image branch runs `z_image_convert`), so the pack carries the plain checkpoint names.

4 · The impact measurement

Hold the FP16 weights, replace **one** layer by its ConvRot INT8 reconstruction in the **original** frame (`dequant(INT8(W·Hᵀ))` = `(q·s)·H`), and rank by the

THE AGGREGATE CONTRIBUTION — TOP 5

#	LAYER	1 − COS
1	**`img_in`**	**1.1247e-05**
2	`controlnet_blocks.58.output_proj`	1.8809e-08
3	`controlnet_blocks.48.output_proj`	1.8321e-08
4	`controlnet_blocks.13.output_proj`	1.8170e-08
5	`controlnet_blocks.21.output_proj`	1.7917e-08

> `img_in` **is 598× above the runner-up, and carries 90.9 % of the summed contribution** (1.1247e-05 of 1.2373e-05; the other 120 layers average 9.39e-09). It is the single entry point converting the encoded control image into the 3072-wide

change in the patch's own output over 8 fixed seeds.

The injection frame is the whole point. The pack stores the **rotated** weight and the kernel rotates the **activation**, so the error reaching the output is `(q·s)·H − W` — the original frame. Injecting in the rotated frame measures the rotation itself, not the error, and inverts the ranking.

> feature all 60 blocks consume, so every block inherits its error. The 60 blocks cross-cancel in the aggregate.

5 · Candidates, and the adopted set

256 ITEMS · SEED 42 · THROUGH THE REAL LOADER

CANDIDATE	INT8	KEPT	COS MEAN	COS M
k=0	121	0	0.9999555935	0.9999507
k=1 *adopted*	**120**	**1**	**0.9999924905**	**0.9999924**
k=5	116	5	0.9999926819	0.9999926

The curve is a single step, not a plateau.

k=0	0.99995079
k=1	0.99999241
k=5	0.99999260
k=10	0.99999284

cos.min · the bar is scaled to k=1

CANDIDATE	INT8	KEPT	COS MEAN	COS M
k=10	111	10	0.9999929108	0.9999928

Protecting `img_in` alone moves the minimum 0.9999507946 → **0.9999924096**: a **597× cut of the error** (3.7e-05 → 6.3e-08) for **+0.18 MB**. k=1 → k=10 buys only +4.2e-07 of the mean for **+80.9 MB** — the gate-satisfied region.

6 · The pack

SOURCE	2161.8 MB (BF16)
PACK	**1083.3 MB** (−49.9 %)
QUANTIZED	120 of 121 (`img_in` kept FP16)
KEPT	241 tensors, all FP16 (Rule 5)
GATE	**PASS**

What the rebuild changed

	PREVIOUS	REBUILT
protected	0	**1 (`img_in`)**
kept dtypes	121 BF16 + 120 F32	**241 FP16**
cos mean	0.9999555935	**0.9999924905**
cos min	0.9999507946	**0.9999924096**
size	1083.80 MB	**1083.28 MB**

> Two defects fixed at once — the missing protection, and the wasted FP32 kept bytes. The rebuilt pack is **0.52 MB smaller** while strictly better.

7 · The acceptance through the real loader

TWO INDEPENDENT HARNESSES · BOTH AGREE

HARNESS	HOW IT BUILDS THE PACK	COS MEAN / MIN (K=1)
`verify_qwen_image_diffsynth_patch.py`	module graph built directly with `mixed_precision_ops`	0.9999924905 / 0.9999924096
`verify_qwen_image_diffsynth_patch_loader.py`	**`HSWQModelPatchLoaderCustom.load_model_patch(name, False)` — the node ComfyUI runs**	**0.9999924905 / 0.9999924096**

The loader harness stages a candidate into `models/model_patches` under a unique name, loads it through the real node, measures, then removes it. The loaded module is confirmed to be `QwenImageBlockWiseControlNet` with 120

INT8 layers and `img_in` kept — so the pack really is the INT8 layout the loader consumes, not a pack that happens to load as FP16.

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8 · Reproduction

```
# the en
#   pyth
#   COMF

# 1 · th
python s
    --pa
    --ou

# 2 · th
python c
    --in
    --ou
    --co

# 3 · th
$env:QDS
python s
    <Con
    impa
```

Determinism

The builder is exact arithmetic over the source tensors and the contribution list, so it is byte-reproducible: re-running step 2 reproduced the placed pack byte for byte (SHA256 `C2899255…C996`, both times).

The acceptance is deterministic too: the 256 inputs come from `torch.Generator("cpu").manual_seed(42 + i)`, and both harnesses returned identical numbers to 10 decimals.

Timing

Contribution ~55 min (121 layers × 9 forwards × 60 blocks, float64). Acceptance ~45 min for 256 items. Builder under a minute.

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9 · Error sources and limits

Error sources in the measurement

- **float32 arithmetic is fatal here.** Residuals run to ~3e5; a float32 dot product over 47M elements accumulates ~1e-3 of noise, larger than the whole signal (< 1.2e-05). Symptom: values all negative or all equal.
- **Concatenating the 60 residuals hides**

Limits of the score

- **Input-distribution dependent.** Measured on fixed random tensors of the correct shape (no VAE / text encoder on this machine). Sound for ranking layers; spot-check once on a real image before release.
- **The patch's own output only.** No DiT, sampler or VAE. Use the trajectory gate when a base model is available.
- **A 256-item set.** `cos.min` is the worst of 256 draws, not a bound. Below ~1e-07 candidates are not resolved — hence k=5 and k=10 tie.
- **Single-layer contributions are marginal.** Many quantized together can partially cancel; acceptances run on the *complete pack*.

Where it does not apply

- Layers that are not 2D Linear (Conv2d, embeddings, norms) stay FP16.
- Widths with no power-of-4 divisor cannot be rotated; they stay FP16.
- A checkpoint whose runtime has no INT8 load path (these need the HSWQ loader).

the ranking. Score per block and average.

- `cond` **must be recomputed inside the loop.** It is produced by `process_input_latent_image`, which runs through `img_in` — itself under test. Precomputing it understates `img_in` by an order of magnitude.
- **The baseline must use the same ops**

class as the pack. Stock is `manual_cast`, INT8 is `mixed_precision_ops`; not equal.

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10 · The experiment ledger

#	WHEN (JST)	WHAT	THE SETTING	COS MEAN / MIN	ADOPTED
1	10-23 11:20	the aggregate contribution	121 layers · 8 seeds · float64 · per-block	—	a ranking
2	10-23 11:33	candidate k=0	121 INT8 · 0 kept	0.9999555935 / 0.9999507946	no
3	**10-23 11:33**	**candidate k=1**	**120 INT8 · 1 kept (`img_in`)**	**0.9999924905 / 0.9999924096**	**yes**

#	WHEN (JST)	WHAT	THE SETTING		COS MEAN / MIN	ADOPTED
4	10-23 11:33	candidate k=5	116 INT8 · 5 kept		0.9999926819 / 0.9999926039	no (size)
5	10-23 11:33	candidate k=10	111 INT8 · 10 kept		0.9999929108 / 0.9999928415	no (size)
6	10-23 11:33	the previous pack	121 INT8 · 0 kept · BF16/F32 kept		0.9999555935 / 0.9999507946	replaced
7	10-23 13:55	the loader acceptance	k=0 & k=1 via `HSWQModelPatchLoaderCustom` · 256 items	k=0: 0.9999555935 / 0.9999507946 · k=1: 0.9999924905 / 0.9999924096		confirms #3
8	10-23 13:00	the reproducibility check	re-run the builder (k=1)		byte-identical SHA256	confirms #3

Every row is a real run on this machine; the raw JSON is kept under `impact/` (git-ignored). One consumer GPU; 256 items per acceptance.

11 · Files

FILE	ROLE
`sdxl/measure_qwen_image_diffsynth_patch_contrib.py`	the criterion (the aggregate contribution)
`clip_convert/convert_qwen_image_diffsynth_patch_impact_convrot_int8.py`	the builder (`--top-k` , FP16 kept)
`sdxl/verify_qwen_image_diffsynth_patch.py`	the acceptance, module graph built directly
`sdxl/verify_qwen_image_diffsynth_patch_loader.py`	the acceptance through the real HSWQ loader
`impact/contrib_qwen_image_canny_diffsynth.json`	the contribution (git-ignored)
`impact/verify_qds_256.json · verify_qds_loader_both.json`	the acceptance results (git-ignored)

Qwen-Image diffsynth ControlNet · ConvRot INT8 model patch · measured 2026-10-23 (JST). The method is stated in `md/How to quantize Text Encoder and ControlNet.md` , section “How a pack is selected”.